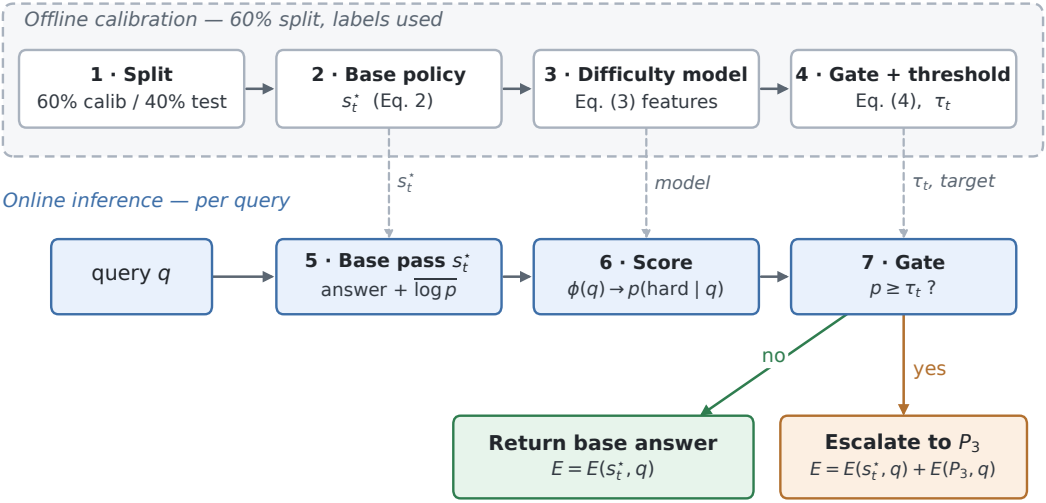
Offline calibration — 60% split, labels used
1 · Split
60% calib / 40% test
2 · Base policy
$s_t^\star$ (Eq. 2)
3 · Difficulty model
Eq. (3) features
4 · Gate + threshold
Eq. (4), τ_t
$s_t^\star$
model
τ_t, target
Online inference — per query
query q
5 · Base pass $s_t^\star$
answer + $\overline{\log p}$
6 · Score
$\phi(q) \to p(\text{hard} \mid q)$
7 · Gate
$p \geq \tau_t$?
no
yes
Return base answer
$E = E(s_t^\star, q)$
Escalate to P_3
$E = E(s_t^\star, q) + E(P_3, q)$